



DO QUANG HUY

Ph.D. Candidate, Kyung Hee University

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[Portfolio Website](#)

EDUCATION

Kyung Hee University

Ph.D. in Electronic Engineering.

Advisor: Prof. Yoon Sang Woong.

Thesis: High Isolation Phase Conjugator and Wideband CMOS Circulator IC for Retrodirective Wireless Power Transfer System.

GPA: 4.0/4.3

Yongin, South Korea

Spring 2021 – Spring 2026

Hanoi University of Science and Technology

B.E. in Mechatronics Engineering.

Advisor: Prof. Vu Toan Thang.

Thesis: Design of an Automatic Balancing System for a Lift-Assist Manipulator.

GPA: 3.2/4.0

Hanoi, Vietnam

Fall 2015 – Fall 2020

SKILLS SUMMARY

- **Technical** : RF/Analog Design, IC Layout, EM Simulation, PCB Design.
- **CAD Tools** : Cadence, Calibre, SpectreRF, ADS, HFSS, SiWave, Altium
- **Programming** : C/C#/C++, Python, Shell.
- **Lab Skills** : Spectrum/Network Analyzer, Signal Generator, Probe Station, Noise Analyzer, Laser Trimming.
- **Soft Skills** : People Management, Scientific Writing, Collaboration.
- **Language** : IELTS 7.0.

WORK EXPERIENCE

Graduate Researcher | [Kyung Hee University](#) **(Full Time)**

Yongin, South Korea

2021.03 – present

RF/Analog IC Design & Tape-Out:

- Design RF/Analog front-end circuits (LNAs, DAs, Mixers, VCOs, Circulators...) and contribute to system architecture for intelligent wireless power transfer.
- Perform schematic design, simulation, layout, and full verification flow (DRC, LVS, PEX).
- Conduct EM simulations (HFSS, Siwave) for on-chip passives and RF structures; prepare design DB for tape-out.

Prototyping, PCB & Testing:

- Develop test plans, design PCB test benches, and perform functional, RF, and system-level measurements.
- Experience with lab equipments: VNAs, Spectrum Analyzers, Oscilloscopes, Signal Generators, Probe Stations, Laser Trimming.
- Debug and optimize designs based on measurement results.

Documentation & Publication:

- Write and contribute to project proposals and research reports.
- Prepare scientific manuscripts, conference papers, and presentations.
- Generate and maintain design documentation including schematics, specs, measurement logs, and test reports.

Hardware Engineer | [VIETMANI JSC](#)
(Full Time)

Hanoi, Vietnam
2019.09 – 2021.02

Control Systems & Structural Design:

- Designed feedback-control system for lift-assisted manipulators using AVR/ARM microcontrollers.
- Performed 3D (NX, AutoCAD) and FEA modeling for structural analysis.

Prototyping & Testing:

- Designed control boards and integrated systems, including sensor/actuator calibration.
- Produced schematics, wiring diagrams, test reports, and user manuals.

Hardware Engineer | [BKAPEMA Co. Ltd.](#)
(Intern)

Hanoi, Vietnam
2018.04 – 2019.08

Control & Automation Systems:

- Designed PLC/HMI automation systems; integrated sensors, relays, and actuators (DC/BLDC/Pneumatic) using embedded controllers.

Documentation & Collaboration:

- Produced wiring diagrams, I/O lists, and operation manuals while ensuring cross-functional system compatibility.

Hardware Engineer | [SELEX MOTORS JSC](#)
(Intern)

Hanoi, Vietnam
2017.06 – 2018.12

Electronics Circuit & PCB Design:

- Designed Battery Management System (BMS) incorporating MCUs, power management circuitry, opto-electrical components, sensors, and analog/mixed-signal components under supervision.

Prototyping & Testing:

- Built and validated prototypes through rigorous execution of test plans and component evaluation.

Documentation and Collaboration:

- Evaluate and select components based on availability, cost, and lifecycle.
 - Collaborated with mechanical and firmware teams on hardware integration and managed Bill of Materials (BOM).
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TEACHING & PEDAGOGY

Teaching Assistant

Assisted in grading and guiding students through advanced circuit design concepts.

- Advanced RFIC Design (Graduate Course) *Spring 2025*

Curriculum Design & Student Mentoring

Designed training curricula and instructed undergraduate students.

- Analog CMOS IC Design. *Summer 2025*
- Fundamentals of Microelectronics; Circuit Analysis. *Winter 2024*
- Basics of Electromagnetics; Semiconductor Physics. *Fall 2024*

Technical Training

Developed and delivered practical training modules on industry-standard tools.

- IC Design Flow: Cadence Virtuoso, Spectre RF, Calibre (TSMC 180nm).
- EM & PCB Design: Ansys HFSS, Altium Designer.
- RF Measurement: VNA, Signal Generators, Spectrum Analyzers.

PROJECTS

IR Receiver Based on Staggered Communication

2025.05 – present

Main Researcher

- Studied staggered commutation techniques for intrinsic image rejection without conventional quadrature mixing.
- Designed and simulated a prototype receiver to evaluate conversion gain, image suppression, noise and linearity performance.

High Isolation Phase Conjugator and Wideband Circulator

2023.03 – 2025.03

Main Researcher

- Designed a phase conjugator integrating Self-Interference Cancellation with a Gilbert-cell mixer to enhance retrodirectivity in wireless power transfer system.
- Studied configurations of an N-path commutated network-based circulator to improve TX-RX isolation and transmission performance.
- Proposed a novel use of non-Foster resonator to widen TX-RX isolation BW.
- Designed a 2.4 GHz magnetic-free circulator.
- Completed tape-out with Samsung 28nm RF CMOS.
- Measurement of phase-conjugator demonstrated state-of-the-art isolation performance, and fabricated circulator achieved 25% fractional bandwidth.

Retro-directive RF SoC with Magnetic-free Circulator

2022.01 – 2023.01

Main Researcher

- Conducted research on magnetic-free circulators for full-duplex communication.
- Designed a fully integrated CMOS circulator including N-path filter-based gyrator and 4-phase clock generator.
- Proposed a negative capacitor (NCAP) delay line to cancel PGD generated by gyrator to extend bandwidth.
- Completed tape-out with Samsung 28nm RF CMOS.

- Fabricated circulator achieved a bandwidth of 65MHz, 13 times greater than that of its narrowband counterpart.

RFIC Design for Retro-directive Beamforming

2021.04 – 2021.12

Research Assistant

- Studied IC design flow and linux-based system management.
- Designed RFIC blocks: Low-noise amplifier (LNA) and drive amplifier (DA).
- Completed tape-out with Samsung 65nm RF CMOS.

PUBLICATION LIST

- **Q. -H. Do**, D. -N. Pham, and S. -W. Yoon, "Wideband CMOS Circulator IC using Non-Foster Elements," (**submitted** to *IEEE Transactions on Microwave Theory and Techniques*)
- **Q. -H. Do**, D. -N. Pham and S. -W. Yoon, "Isolation-Enhanced Phase Conjugator Design for Retro-Directive Wireless Power Transfer System," in *IEEE Microwave and Wireless Technology Letters*, doi: 10.1109/LMWT.2025.3597259.
- **Q. -H. Do**, T. -B. Ngo, and S. -W. Yoon, "An Ultra-compact Fully Integrated Circulator in 28 nm CMOS IC Technology," *Journal of Semiconductor Technology and Science*, doi: 10.5573/JSTS.2025.25.2.184.
- **Q. -H. Do**, T. -B. Ngo, and S. -W. Yoon, "A CMOS Low-pass Filter with Group Delay Cancellation using Non-Foster Element Circuits," *Journal of Semiconductor Technology and Science*, doi: 10.5573/JSTS.2024.24.2.105.
- T. -B. Ngo, **Q. -H. Do**, and S. -W. Yoon, "Leakage Power Canceling Module with a Negative Capacitor for a Circulator's Isolation," *Journal of Electromagnetic Engineering and Science*, doi: 10.26866/jees.2023.2.r.151.
- T. -B. Ngo, **Q. -H. Do**, and S. -W. Yoon, "A Wideband Circulator Leakage Canceler for Retro-Directive RF System," in *IEEE Microwave and Wireless Components Letters*, doi: 10.1109/LMWC.2022.3170590.
- T. -B. Ngo, **Q. -H. Do**, and S. -W. Yoon, "A Compact 6-bit Phase Shifter in 65 nm RF CMOS Technology for ISM Band," *Journal of Semiconductor Technology and Science*, doi: 10.5573/JSTS.2022.22.3.198.

REFERENCES

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